

Sungok Woo

Full-Stack & Systems Developer

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PROFILE

Computer Science student at UC Irvine who works across the full stack and down to the metal. I have shipped a production tutoring platform spanning web and mobile, integrated the Anthropic Claude API for AI matching and report generation, and built systems-level software in C including a UNIX shell, a custom memory allocator, and a multi-threaded TCP server.

EXPERIENCE

Lead Developer / Growth Tutoring Platform Nov 2025 - Present

- Built a full-stack tutoring platform across web and mobile (React Native/Expo) with real-time messaging, Stripe payment processing, and Amazon Chime SDK video sessions.
- Engineered an AI tutor matching system using the Anthropic Claude API, ranking tutors against student intake profiles and returning match scores with natural-language explanations.
- Developed an AI Progress Report feature where Claude generates professional parent-facing session narratives from tutor inputs, with subscription-based gating and automated reward detection.
- Advanced to the semi-finals of the Stella Zhang New Venture Competition and Butterworth Competition, presenting the platform's AI ecosystem and business model to investors and technical judges.

React · Spring Boot · MySQL · React Native · Anthropic Claude API · Stripe · AWS

PROJECTS

Information Retrieval Search Engine Jan 2026 - Mar 2026

- Built a multithreaded web crawler targeting UCI academic domains with politeness enforcement, trap detection, and near-duplicate filtering.
- Developed a disk-based inverted index with Porter stemming, bigram indexing, and important tag-boosting.
- Implemented TF-IDF search with cosine normalization, bigram query boosting, and URL penalties.

Python · Flask · NLTK · BeautifulSoup · Tkinter · JSON

Low-Level System Programming Sep 2025 - Dec 2025

- Built a UNIX shell supporting multi-stage pipelines, I/O redirection, background job control, and signal handling using fork/execvp.
- Implemented a custom memory allocator using segregated free lists, boundary-tag coalescing, and block splitting.
- Designed a multi-threaded TCP server across three concurrency models: mutex threading, reader-writer locks, and producer-consumer with semaphores.

C · UNIX · Pthreads · TCP Sockets · Valgrind · GDB · Make

EDUCATION

UC Irvine

B.S. Computer Science
Sep 2024 - Sep 2026 (exp.)

Orange Coast College

Computer Science
Aug 2022 - May 2024

SKILLS

Languages

C / C++ Java Python JavaScript SQL
HTML / CSS

Frameworks

React React Native Node.js Spring Boot
Flask Bootstrap

Tools & Platforms

Git MySQL AWS Claude API Tomcat
Maven

COURSEWORK

Machine Learning · Algorithm Analysis · Information Retrieval · Data Management (SQL) · Computer Organization · Quantum Computing · Data Structures

LANGUAGES

Korean	Native
English	Professional
French	Intermediate